

VISHAL SINGH

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Software Engineer experienced in backend systems and AI-powered automation. Strong foundation in Python, scalable application development, cloud technologies, and problem-solving with **870+** LeetCode problems solved.

SKILLS

- **AI & LLM Engineering:** LangGraph, Agentic Workflows, LLM APIs, Prompt Engineering, LLM Evaluation, Structured Output Generation, Embeddings, OCR, Document Intelligence
- **Programming & Backend:** Python, C++, JavaScript, TypeScript, Node.js, Express.js, FastAPI, REST APIs
- **Frontend & Web:** React.js, Next.js, Vite, HTML, CSS, Tailwind CSS
- **Databases & Data:** SQL, MySQL, PostgreSQL, MongoDB, Supabase, Firebase, Database Indexing, Query Optimization, Data Modeling
- **Cloud & DevOps:** AWS (EC2, IAM, Route 53), Git, GitHub Actions, CI/CD, Vercel, Deployment Automation
- **Software Engineering:** Data Structures & Algorithms, OOP, Design Patterns, System Design

CERTIFICATIONS & ACHIEVEMENTS

- **870+ LeetCode Problems Solved | 600+ Day Coding Streak** | [LeetCode](#)
- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner

EXPERIENCE

AI Software Engineer | Accenture

On-site | Oct 2025 - Present

- Architected a **Python/LangGraph** agentic orchestration framework transforming SAP Functional Documents into validated **CSN v2.0 JSON**, processing **500+ documents** and cutting manual data-modeling effort by **8+ hours/sprint**.
- Built an automated entity-extraction and relationship-mapping pipeline achieving **95%+ schema-compliant accuracy**, backed by **guardrail workflows (LLM-as-a-Judge** scoring, confidence thresholds, human-in-the-loop review) that improved output reliability across downstream systems.
- Integrated generated models into the Functional Document → JSON → SAP Datasphere workflow, enabling automated retail analytics across customer, product, store, and transaction domains.

Backend Engineer Intern | CEIAmerica

Internship | Sep 2023 - Oct 2023

- Built a secure authentication system that hardened access control and closed unauthorized-access gaps across internal tooling.
- Improved SQL query performance by **25%** through optimized indexing and refined JDBC connection handling on high-traffic queries.
- Developed backend services powering three internal applications, reducing manual intervention for the operations team.

PROJECTS

ExamChecker | AI Handwritten Answer Evaluation Platform [*Node.js, React, LLM, PDF Generation*] - [\[Live App\]](#) *Jan 2025 - Mar 2025*

- Built an AI-powered evaluation platform cutting grading time from **20 minutes** to under **2 minutes** per paper via automated extraction, grading, and report generation.
- Engineered a single-call multimodal grading workflow using **Gemini API** on a browser-based video-to-document pipeline (motion detection, frame-stability analysis), identifying and evaluating scanned answers while cutting API calls, latency, and token costs vs. page-wise processing.
- Designed a resilient grading engine with multi-level fallback (Gemini, sentence-transformer embeddings, keyword-similarity scoring), supporting offline-first operation and automated PDF reports.

Aptitude Web App | Quiz & Assessment Platform [*MERN Stack, AWS EC2, MongoDB*]

Jun 2024 - Jul 2024

- Built and deployed a full-stack assessment platform with **500+** questions, authentication, and analytics, increasing user engagement by **40%**.
- Reduced query latency by **30%** through MongoDB indexing, schema optimization, and efficient API design.
- Improved application performance by **75%** by migrating to **AWS EC2** with dedicated compute, configuring **Route 53, Elastic IPs, AMI backups, and EBS snapshots** for disaster recovery and reliability.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science Engineering, CGPA: 8.4

Tamil Nadu, India

2021 – 2025